

Project requirements

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level : simple

Team Size : 4

Stock Market Analysis System (Using Streamlit)

◆ 1. Project Overview

This project is a web-based application developed using Streamlit that allows users to view and analyze stock market data. The system provides simple visualizations and basic stock information to help users understand market trends.

◆ 2. Objectives

- To understand how stock market data is collected and displayed
- To build a simple interactive web application
- To visualize stock price trends using charts
- To provide basic stock information to users

◆ 3. Scope

The system will:

- Allow users to search for stock symbols
- Display current and historical stock prices
- Show graphical representation of stock trends
- Provide a simple and user-friendly interface

◆ 4. Functional Requirements

The application should:

- Accept stock symbol input from the user
- Fetch stock data from an API (e.g., Yahoo Finance)
- Display:
 - Stock name and current price
 - Historical data (last 7 days / 1 month)

- Generate line charts for price trends
- Handle invalid inputs with error messages

◆ 5. Non-Functional Requirements

- The system should be easy to use
- The interface should be simple and clear
- Data should load quickly
- The application should be reliable and responsive

◆ 6. Tools and Technologies

- **Frontend & Backend:** Streamlit
- **Programming Language:** Python
- **Libraries:**
 - Pandas (data handling)
 - Matplotlib or Plotly (visualization)
 - yfinance (stock data API)

◆ 7. System Workflow

1. User enters a stock symbol (e.g., AAPL)
2. The system retrieves stock data from the API
3. The data is processed and displayed
4. Charts are generated to show trends

◆ 8. User Interface Requirements

- Input field for stock symbol
- Display area for stock data
- Chart section for visualization
- Clean and simple layout

◆ 9. Testing Requirements

- Test with valid stock symbols
- Test with invalid inputs
- Verify chart accuracy
- Check system performance

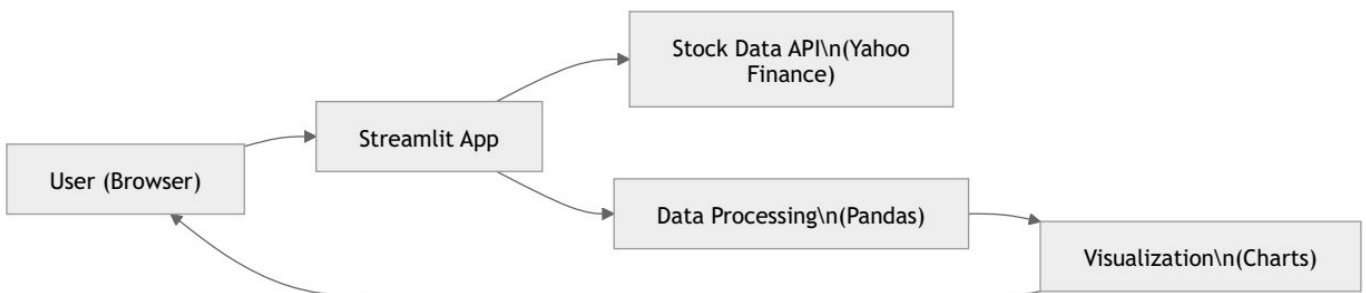
◆ 10. Expected Output

- Display of stock price information
- Graph showing stock trends
- Simple and interactive web interface

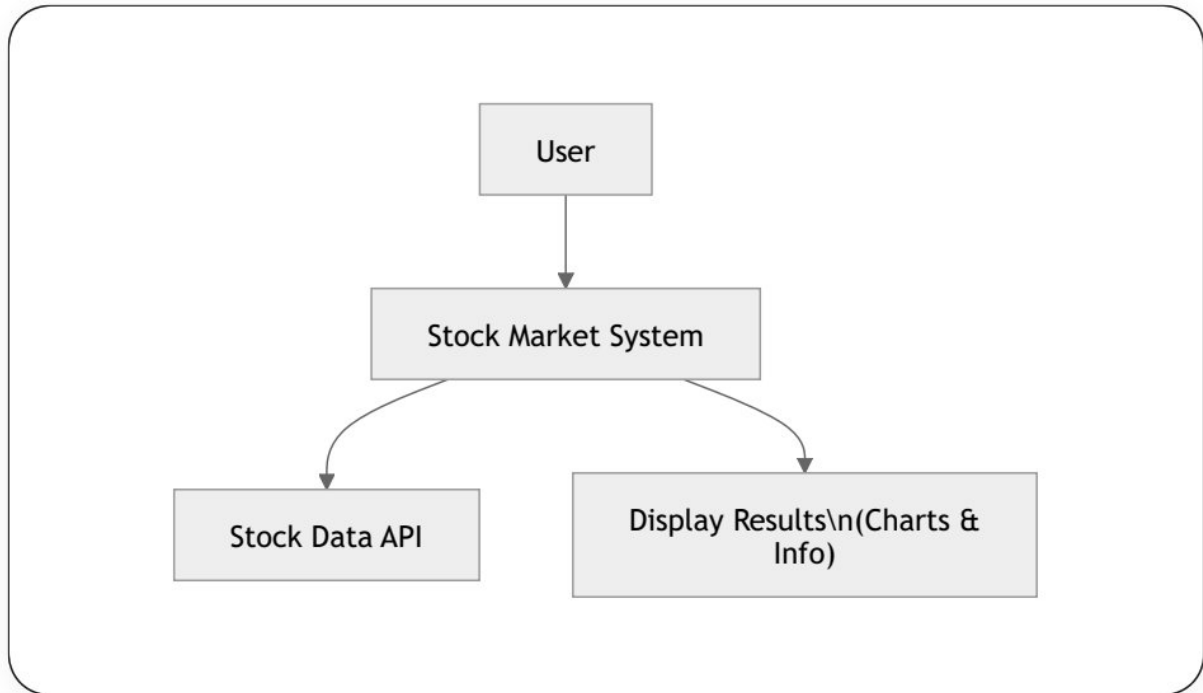
◆ 11. Conclusion

This project demonstrates basic knowledge of stock market data analysis and web application development using Streamlit. It is suitable for beginners and provides a foundation for more advanced systems.

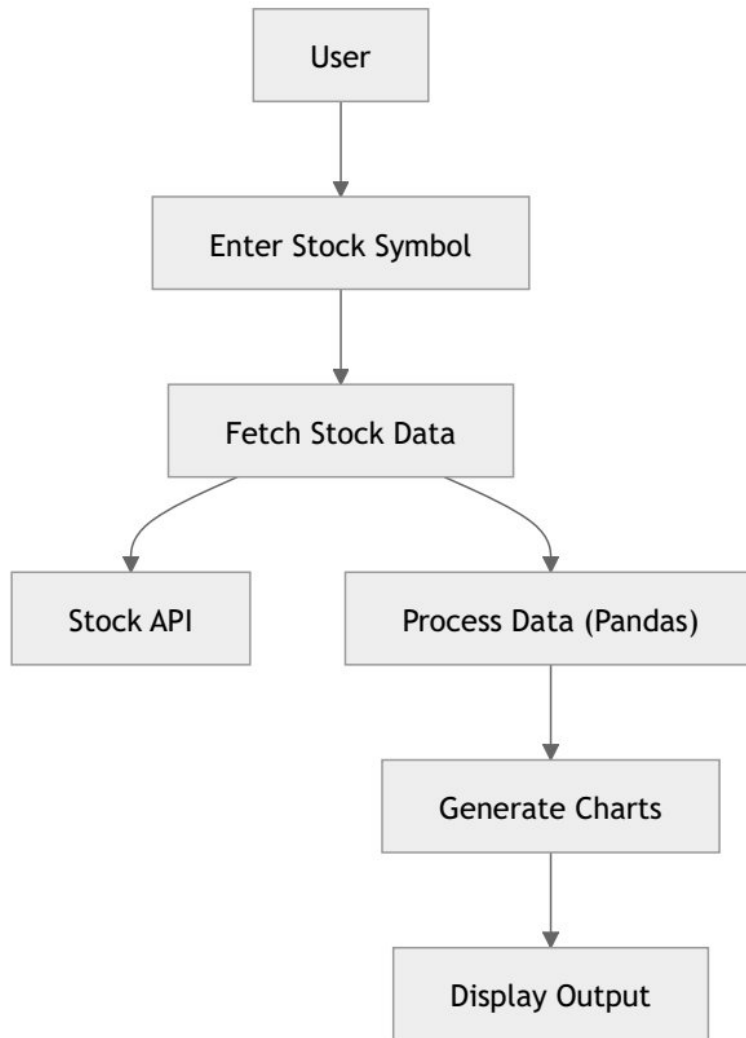
1. System Architecture Diagram (Image Style)



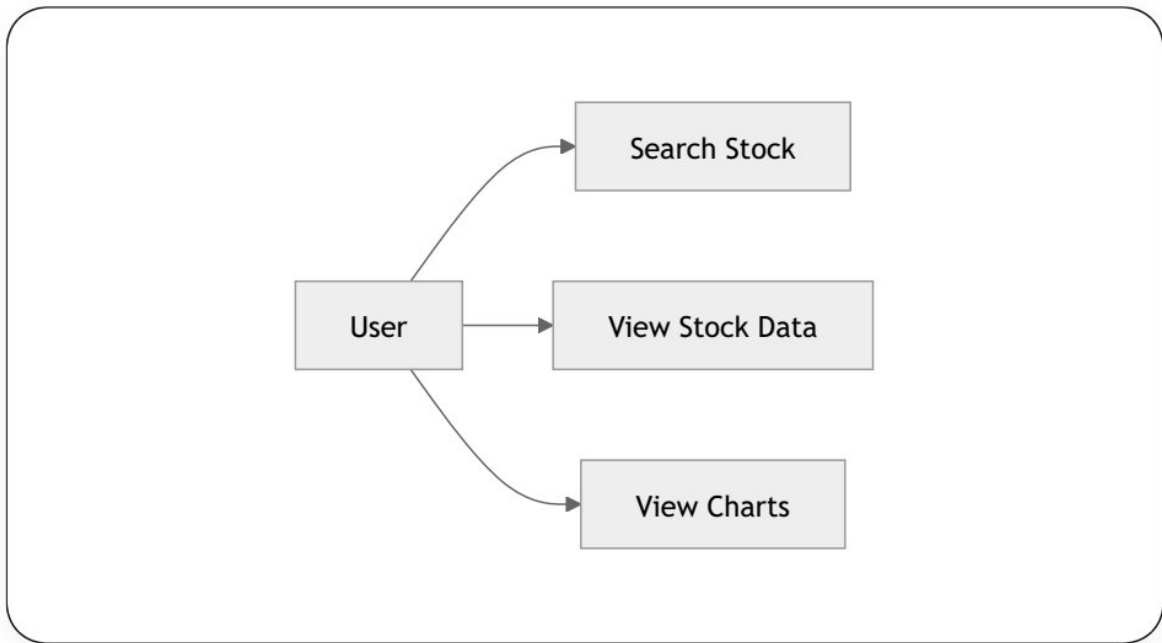
2. Data Flow Diagram (DFD Level 0)



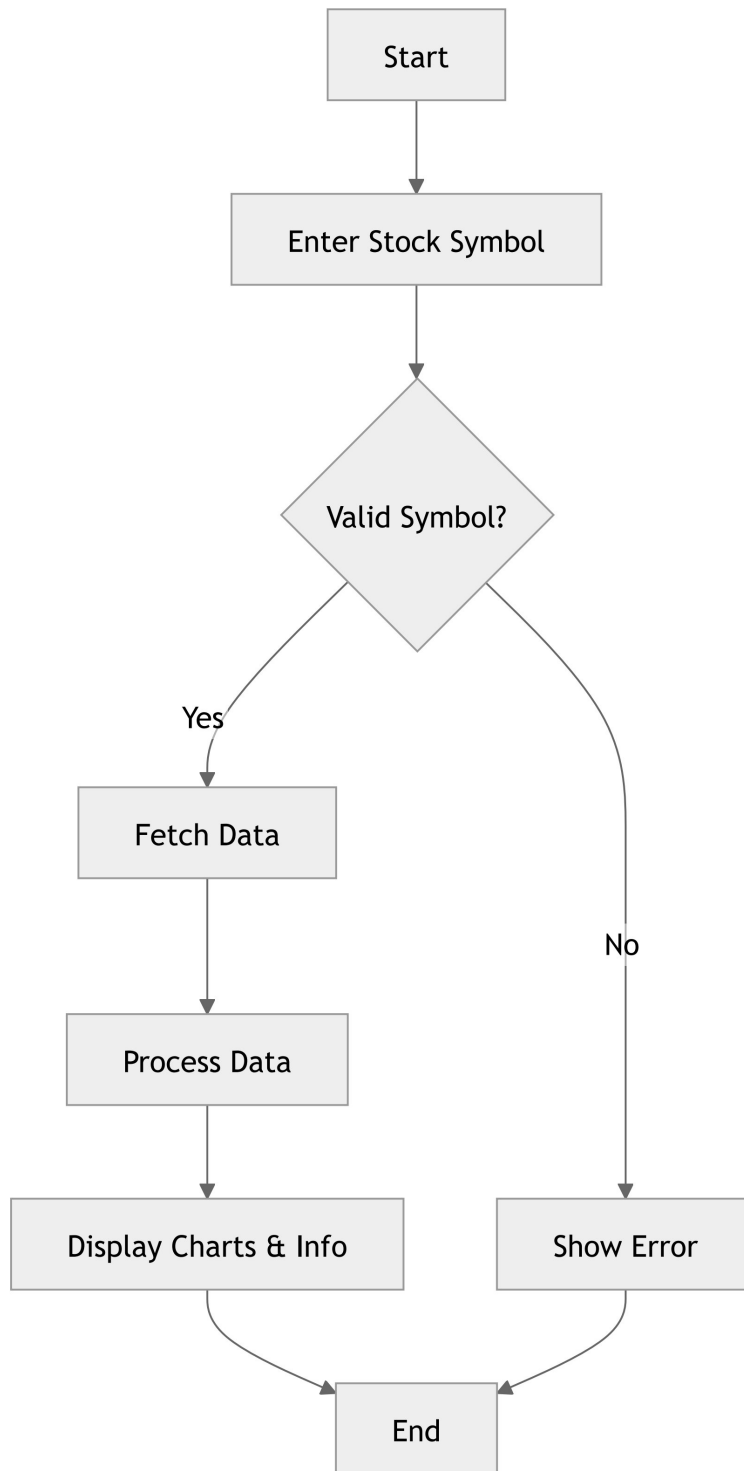
3. Data Flow Diagram (DFD Level 1)



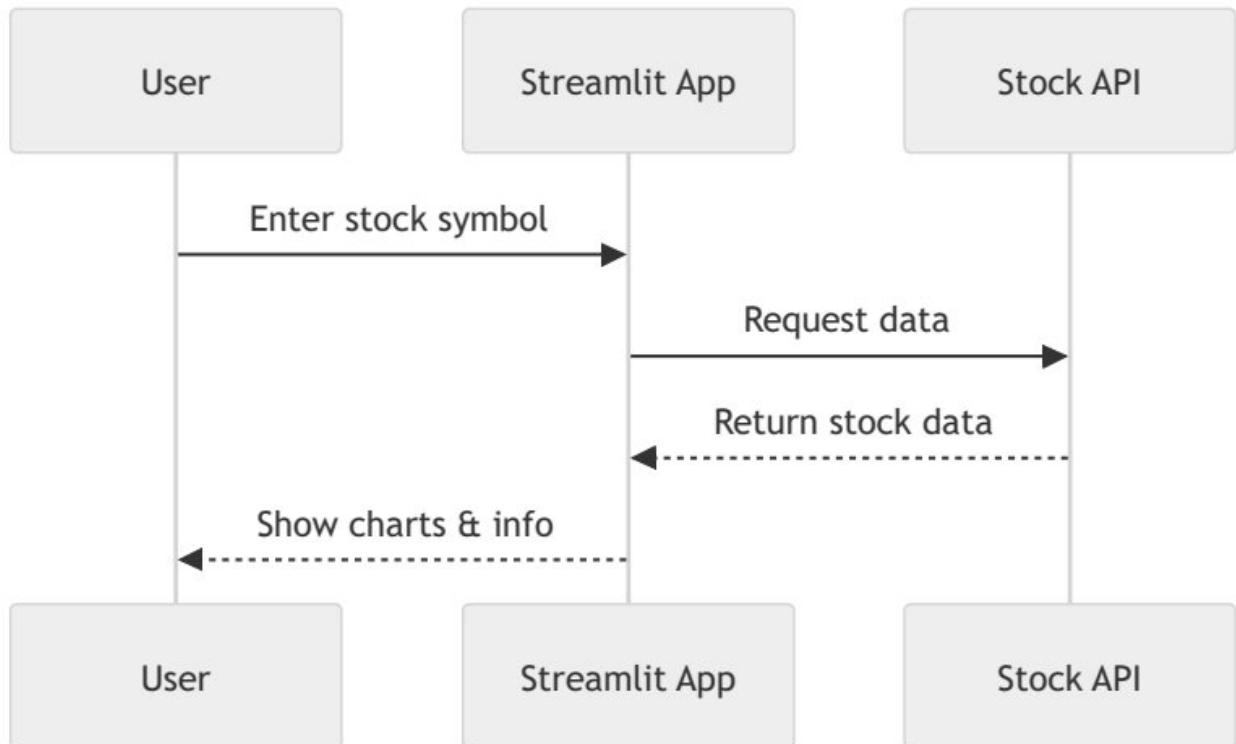
4. Use Case Diagram



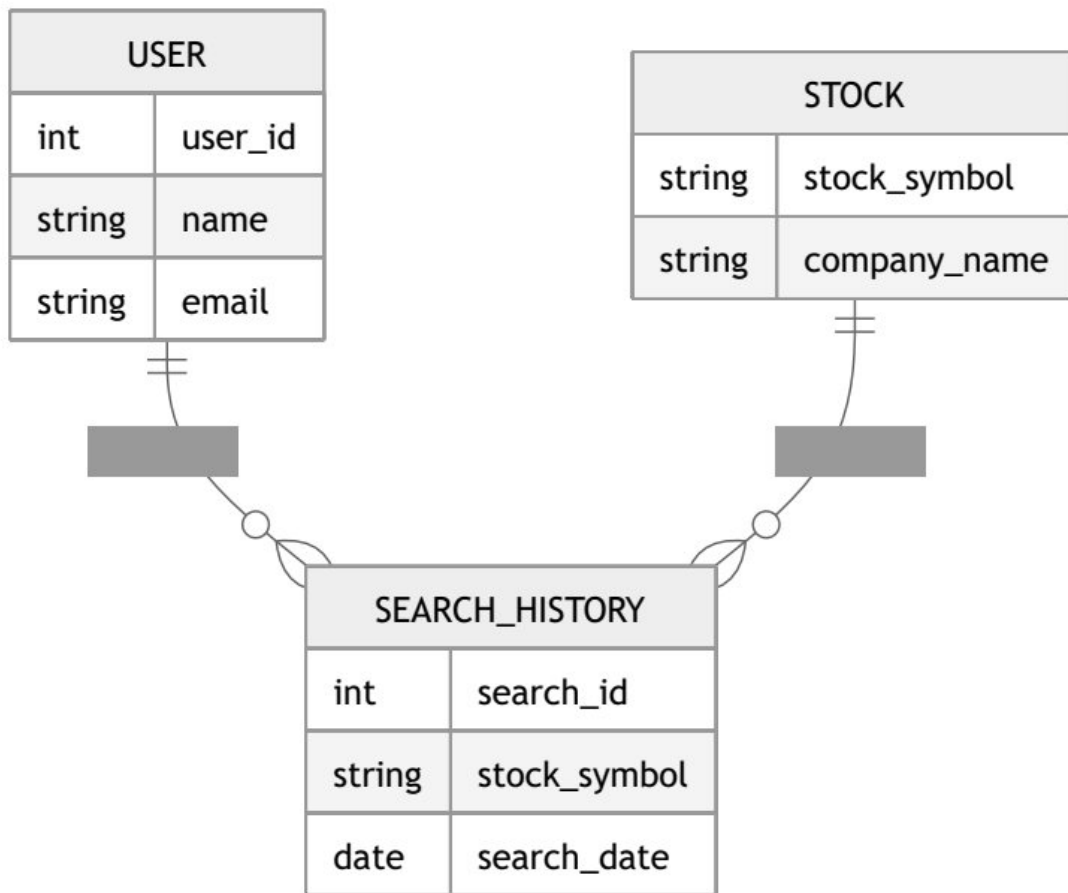
5. Activity Diagram



6. Sequence Diagram



7. ER Diagram (Database – Simple)



Team Task Allocation

Member 1 – Project Lead / Backend Developer

Main Role: API Integration + Core Logic

- Set up Python project structure
- Integrate stock data API using **yFinance**
- Handle stock symbol input and validation
- Fetch current + historical stock data
- Manage error handling (invalid symbols, API failures)

Member 2 – Data Processing & Analysis

Main Role: Data Handling

- Clean and structure stock data using **Pandas**
- Extract required fields (open, close, high, low, volume)
- Prepare data for visualization
- Compute simple indicators (optional): moving average, daily change

Member 3 – Visualization Developer

Main Role: Charts & Graphs

- Create stock trend charts using **Matplotlib or Plotly**
- Build interactive visualizations (line charts, trends)
- Design graph layout for clarity
- Ensure charts update dynamically based on user input

Member 4 – UI Developer / Tester / Documentation

Main Role: Streamlit UI + Testing

- Build Streamlit interface (input box, buttons, layout)
- Display stock info and charts in UI
- Improve user experience (clean design)
- Test application with valid/invalid inputs
- Prepare documentation (report, screenshots, PPT if needed)